Week 8 Recitation

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ECON2001.01 (Prin. Micro.)

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Chap 7

Revisit Overview – Market Efficiency

Absent of a **market distortion**, equilibrium in a perfectly competitive market is **pareto efficient**

- Social surplus is maximized

The equilibrium price has **three distinct functions** in a perfectly competitive market: (Invisible Hands)

1. Efficiently allocates resources across buyers and sellers
   - To max Social surplus, matches buyers with highest reservation value to seller with lowest reservation value
2. Efficiently allocates resources across firms
   - MC $=$ MR $\Rightarrow$ All firms produce their last unit at the same cost (i.e., **MCs are all equalized** by the efficient allocation)
3. Efficiently allocates resources across markets
   - LR equilibrium, all firms earn zero profit (No other outside options for firms that yield a greater return)
   - Price guides resources for the firms to produce at the lowest cost

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Revisit – Across Buyers and Sellers

About the point (1):

Matching buyers and sellers regarding their reservation values $\Longrightarrow$ Social Surplus is maximized?

Exhibit 7.1 Reservation Values of Buyers and Sellers in the iPhone Market

Quantities are only equal at $40

Buyers	**Res. Value ($)**	**Cum. Q**	**Sellers**	**Res. Value ($)**	**Cum. Q**
Madeline	$70	1	Tom	$10	1
Katie	$60	2	Mary	$20	2
Sean	$50	3	Jeff	$30	3
Dave	$40	4	Phil	$40	4
Ian	$30	5	Adam	$50	5
Kim	$20	6	Matt	$60	6
Ty	$10	7	Fiona	$70	7

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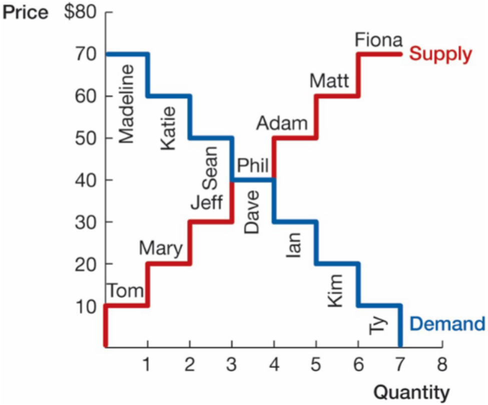
Revisit – Across Buyers and Sellers

About the point (1):

Matching buyers and sellers regarding their reservation values
$\Longrightarrow$ Social Surplus is maximized?



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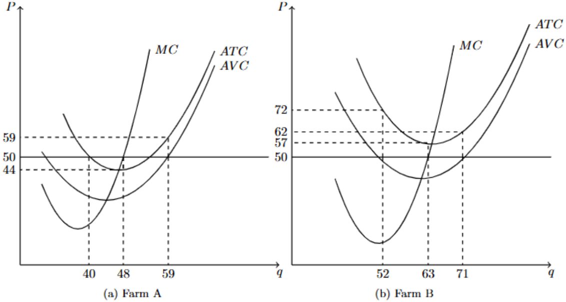
Revisit – Across Buyers and Sellers

Figure 2: **Smith and Children's Corn Farms**



(a) Refer to the above figure (quantity in bushels of corn). Suppose the price of a bushel of corn is $50. As the manager of the two farms, what is Smith and Children profit maximizing output at each farm? Calculate Smith and Children Corn Farms profit?

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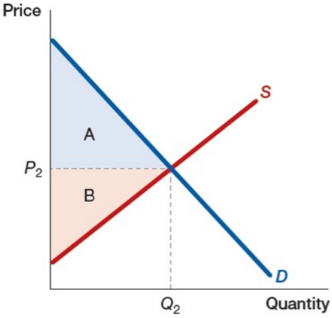
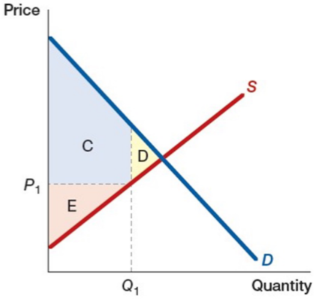
Revisit – Dead Weight Loss

What happens when market intervention?



(a) Surplus in a Free Market



(b) Surplus with a Price Control

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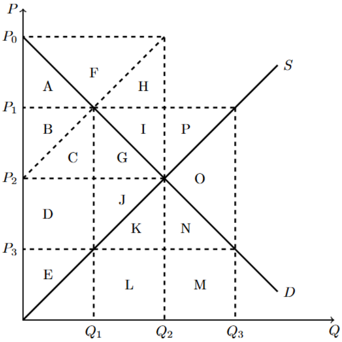
Revisit – Dead Weight Loss

What happens when market intervention?

2. The production of certain goods can discharge pollutants into the air, waterways, and soils we inhabit, cultivate, and consume. The emission of such pollutants bears external costs to society. For example, when people breathe certain pollutants, they can become ill, causing a person to stay home until they feel better. When someone misses work, they forego wages. The loss of income from staying home resulting from an illness borne from pollutants is an example of an external cost. Production decisions do not account for these external costs. By not accounting for these external costs, firms expand production beyond what is Pareto efficient.

Figure 1: **Market for ABC Chemicals**



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Revisit – Dead Weight Loss

What happens when market intervention?

Table 1: **Market Surpluses**

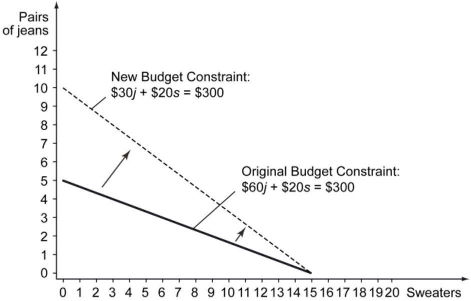
Socially Efficient Outcome		Market Outcome	
Surplus	Value	Surplus	Value
Consumer Surplus		Consumer Surplus	
Producer Surplus		Producer Surplus	
Pollution Costs	$C + D + E$	Pollution Costs	$C + D + E + G + H + I + J$
Social Surplus		Social Surplus	
		DWL	

Review Chap 5

Revisit – Quiz

2) Now suppose your income returns to its original level, \$300, the price of sweaters remains constant at \$20, but the price of jeans falls from \$60 per pair to \$30 per pair, so your new budget constraint is $\$30j + \$20s = \$300$. This rotation of your budget constraint is shown in the graph below.

2) ________



As a result of the decrease in the price of jeans from \$60 per pair to \$30 per pair, the opportunity cost of a pair of jeans ________.

A) remains unchanged at 3 sweaters

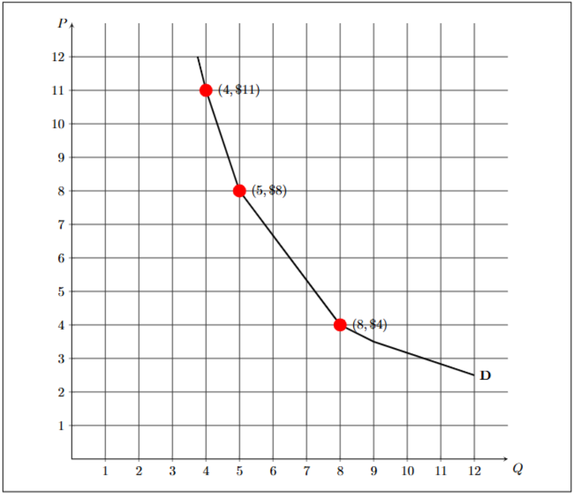
B) falls to 2 sweaters

C) falls to $\frac{2}{3}$ sweaters

D) falls to 1.5 sweaters

Review Chap 5

Revisit – Quiz



(c) Derive the price arc-elasticity of demand for Good A when the price of Good A decreases from $11 to $4, and explain how responsive the buyer is to this price change. Show all work and round to four decimal places if necessary.

Review Chap 6

Revisit – Quiz

5) If a firm is maximizing profits in the short run at a level of output where Average total cost > Price > Average variable cost, then the firm is ________. 5) ________

A) earning positive economic profits

B) realizing losses and will minimize losses by shutting down

C) realizing losses but minimizing losses by continuing to produce

D) breaking even (zero economic profits)